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vovuh's blog

Codeforces Round #560 (Div. 3) Editorial

 By [vovuh](#), [history](#), 7 years ago,

1165A - Remainder

 Idea: [vovuh](#)
[Tutorial](#)
[Solution](#)

1165B - Polycarp Training

 Idea: [MikeMirzayanov](#)
[Tutorial](#)
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1165C - Good String

 Idea: [BledDest](#)
[Tutorial](#)
[Solution](#)

1165D - Almost All Divisors

 Idea: [MikeMirzayanov](#)
[Tutorial](#)

1165D - Almost All Divisors

Suppose the given list of divisors is a list of **almost all** divisors of some x (in other words, suppose that the answer exists). Then the minimum divisor multiplied by maximum divisor should be x . This is true because if we have a divisor i we also have a divisor $\frac{n}{i}$. Let's sort all divisors and let $x = d_1 \cdot d_n$.

Now we need to check if all divisors of x except 1 and x are the permutation of the array d (check that our answer is really correct). We can find all divisors of x in $O(\sqrt{x})$, sort them and compare with the array d . If arrays are equal then the answer is x otherwise the answer is -1.

[Solution](#)

1165E - Two Arrays and Sum of Functions

 Idea: [vovuh](#)
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1165F1 - Microtransactions (easy version)

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Before contest
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Idea: **BledDest**[Tutorial](#)[Solution](#)

1165F2 - Microtransactions (hard version)

Idea: **BledDest**[Tutorial](#)[Solution](#)

Tutorial of Codeforces Round 560 (Div. 3)



codeforces, 560, third division, editorial



+32



vovuh



7 years ago



70



Comments (30)

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rupav

7 years ago, [hide](#) # | ☆

Still can't understand problem E. Why we take in reverse order array b ?

→ [Reply](#)

▲ -6 ▼



burt_macklin_

7 years ago, [hide](#) # | ☆

I was hoping that editorial will give a proof for greedy strategy in problem C. :(

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▲ 0 ▼

$$\begin{matrix} k_1 & k_2 \\ b_1 & b_2 \end{matrix}$$

kkbb

7 years ago, [hide](#) # ▲ | ☆

But it's trivial

→ [Reply](#)

▲ 0 ▼



burt_macklin_

7 years ago, [hide](#) # ▲ | ☆

I don't know. The one I know is very very complex and I am not even sure if it is correct. Could you please tell yours?

→ [Reply](#)

▲ 0 ▼

$$\begin{matrix} k_1 & k_2 \\ b_1 & b_2 \end{matrix}$$

kkbb

7 years ago, [hide](#) # ▲ | ☆

Consider the left-most pair that violates the condition. To fix this, we either remove one character from the pair, or one character to the left of the pair. Notice that whichever we do, the effect on the characters to the right of the pair is the same. So, we can always just remove one character from the pair.

→ [Reply](#)

▲ +4 ▼



Kya_Hi_Farak_Padhta_Hai

7 years ago, [hide](#) # ▲ | ☆ ▲ +1 ▼

Hello can you please tell that in E question if first i were to multiply $\min a[i]$ with its corresponding $\max b[i]$ and then apply modulo function how would i do it? Like i want to do first: $c[i] = a[i] * b[i]$ and then $c[i] * (n-i) * (i+1)$. What is the

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correct approach or doing modulo?

→ [Reply](#)



_orcus

7 years ago, [hide](#) <#> | ☆

Is there any way to prove why a greedy solution works in problem C?

→ [Reply](#)

▲ 0 ▼



burt_macklin_

7 years ago, [hide](#) <#> ^ | ☆

<https://codeforces.com/blog/entry/67041?#comment-511546>

→ [Reply](#)

▲ 0 ▼



irlbeck2

7 years ago, [hide](#) <#> | ☆

In Problem E, how do we derive the formula for the number of times $a_i * b_i$ will appear in the final answer?

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▲ 0 ▼



_void

7 years ago, [hide](#) <#> ^ | ☆

← Rev. 3

▲ +1 ▼

This is the number of l and r such that a_l and b_r are going to contribute to $f(l, r)$. There are i possible left borders $(1, 2, \dots, i)$, and $n-i+1$ right borders $(i, i+1, \dots, n)$. We can choose those borders independently, so the number of ways of doing this is $i \cdot (n-i+1)$ according to the product rule.

→ [Reply](#)

7 years ago, [hide](#) <#> | ☆

I have a question on F1. So far what I understand is that you buy the microtransactions on the last day of their sales, unless you can buy out all of the items on the current day. However I am confused since the last day of the sale might be far away, and it would be more optimal to use the current sale.

▲ 0 ▼



vroooooom

For instance, if you have only have 1 item, and 7 microtransactions for it, with sales on day 5 and 10, this algorithm will not do anything until day 10, where it will buy all of the microtransactions that it needs to, and finish off on day 10.

However, you could just spend 5 bourles on day 5, and then buy two transactions on day 9, to finish off the purchases on day 9, which occurs before day 10.

Obviously either my understanding of the problem or tutorial is flawed and I would greatly appreciate it if someone could explain my mistake to me.

→ [Reply](#)



sankalp_

7 years ago, [hide](#) <#> ^ | ☆

Binary search succeeds for day = 10, so we search over the range 1 to 9 and finally arrive at 9 to be the minimum day I think.

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▲ 0 ▼



Pj18

7 years ago, [hide](#) <#> | ☆

In F the question doesn't ask us to keep the money spent minimum, so why don't we buy all type on the first day?

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▲ 0 ▼



verymuch4470

7 years ago, [hide](#) <#> ^ | ☆

you need to have enough money

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▲ 0 ▼



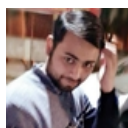
FITHOUk17

7 years ago, [hide](#) <#> | ☆

← Rev. 2 ▲ 0 ▼

In problem E, I still don't understand why the value $(a_i * b_i)$ occur $i * (n - i + 1)$ times in the answer!

Can someone explain it to me :(?

→ [Reply](#)

iabhishek15

6 years ago, [hide](#) <#> ^ | ☆

← Rev. 3 ▲ 0 ▼

your answer = <https://codeforces.com/blog/entry/67041?comment=512045>

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Sam_Jn

7 years ago, [hide](#) <#> | ☆

▲ 0 ▼

Problem D(w.r.t. tutorial): why is it necessary to check that x is the correct answer?

→ [Reply](#)

eng_hardik

7 years ago, [hide](#) <#> | ☆

▲ 0 ▼

Getting TLE on test case 10 in [problem E](#) ... My [solution](#) is in JAVA ... Anybody please help me out

→ [Reply](#)

suhail25

7 years ago, [hide](#) <#> ^ | ☆

▲ 0 ▼

It's bec of the Arrays.sort (dual pivot quicksort algorithm which has n^2 in worst case scenario), so must use mergesort or collection library. It happened with me too :([here](#) you can see.

→ [Reply](#)

eng_hardik

7 years ago, [hide](#) <#> ^ | ☆

▲ 0 ▼

Thanks

→ [Reply](#)

ZxCv

7 years ago, [hide](#) <#> | ☆

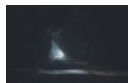
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In problem E, why are we taking the extra term $i*(n-i-1)$?

→ [Reply](#)7 years ago, [hide](#) <#> ^ | ☆

▲ 0 ▼

Because it is a double summation. When you'll simplify it, you'll get that term.



_trojan_horse

Think like: The no. of sequences (l, r) of which $(a_i * b_i)$ is a part .

'l' can be chosen in 'l' no. of ways whereas 'r' can be chosen in 'n-i+1' no. of ways (all possible combinations are valid). So each term will have contribution $i*(n-i+1)$.

→ [Reply](#)

2022OVO

7 years ago, [hide](#) <#> | ☆

▲ 0 ▼

I still don't understand why the solution of D use vector<pair<long long, int>> val(n). What's the use of val[i].second here?

→ [Reply](#)

pr3pony

7 years ago, [hide](#) <#> | ☆

▲ 0 ▼

F can be solved in $O(n + m \log m)$ using BST or segment tree [54494060](#) . The time complexity is independent of the range of $d_j, k_i, sum(k)$ if they can be stored in 64 bits integers.



... came up with the solution because I didn't see the sentence "sum of all m_i is not less than 1 and not greater than $2 \cdot 10^5$ " at first.

→ [Reply](#)



Errichto

7 years ago, [hide](#) <#> | ☆

▲ +1 ▼

I explained problem E in detail in my Youtube video, enjoy. <https://www.youtube.com/watch?v=ThjkHw6vxv8>

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aman_naughty

7 years ago, [hide](#) <#> | ☆

▲ 0 ▼

for d why can't the answer be the lcm of given divisors??

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minhnguyent546

4 years ago, [hide](#) <#> ^ | ☆

▲ 0 ▼

Of course we can. But there is a special case when $X = p * p$, then divisors array = {p}, in this case just take lcm = p^2 .

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kaneki_007

7 years ago, [hide](#) <#> | ☆

▲ 0 ▼

Can anyone help me with problem e bcoz i am not understanding after the first summation of $a[i]*b[i]$ i.e. summation(f(l,r)) how did we get $val[i]=a[i]*i*(n-1-i)$?

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aad3shk

6 years ago, [hide](#) <#> | ☆

← Rev. 3 ▲ 0 ▼

Hello, i have a question for Problem D: In test case #6: 13 802 5 401 4010 62 2 12431 62155 310 10 31 2005 24862 the expected answer is : -1, when the correct answer should be 124310, could someone explain what am i missing ? Thanks

→ [Reply](#)



Imperial_Lord

6 years ago, [hide](#) <#> ^ | ☆

▲ 0 ▼

The answer -1 is correct. If the value for n is odd, it means that the number can be represented as $d[n/2] * d[n/2]$, where d is its divisors array. Thus we need to add the $d[n/2]$ element again in the array, and then sort it. In this case, the added element will be 401, which can't be true, because $401 * 401$ is not equal to $2 * 62155$. So the result is -1

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